CHURNPREDICT | TECHNICAL & BUSINESS AUDIT REPORT

1. Project Overview

ChurnPredict is an end-to-end intelligence platform designed to identify and mitigate customer attrition (churn). Built for subscription-based businesses, it leverages Machine Learning to transform raw customer data into actionable retention strategies. The application combines a high-performance FastAPI backend with an interactive Streamlit dashboard to provide real-time risk scoring and business analytics.

2. Problem Statement

In the modern SaaS and Telecom landscape, customer acquisition costs are 5-25x higher than retention. **Customer churn**—when a user stops using a service—directly impacts monthly recurring revenue (MRR) and long-term viability.

- **The Problem:** Companies often react *after* a customer has already left.
- **The Solution:** ChurnPredict enables proactive intervention by identifying "High Risk" customers before they churn, allowing teams to offer targeted incentives and improve satisfaction.

3. System Workflow (Step-by-Step)

1. **Secure Entry:** Users authenticate via a high-contrast, structured login card.
2. **Dashboard Access:** Upon login, the system initializes session state and navigates to the intelligence panel.
3. **Data Ingestion:** Administrators upload customer datasets (CSV) through the sidebar.
4. **Feature Mapping:** The system dynamically identifies columns for Churn, Tenure, Monthly Charges, and Contract types to ensure compatibility with diverse data structures.

5. **Preprocessing Engine:** Data is automatically cleaned, missing values are imputed, and categorical variables are encoded.
6. **AI Analysis:** The trained ML model processes the features to generate a **Churn Probability** (0% to 100%).
7. **Risk Classification:** Customers are classified into **Low, Medium, or High Risk** based on probability thresholds.
8. **Visualization:** Real-time KPIs and interactive charts (impact factors, tenure trends, contract distribution) are rendered.
9. **Strategic Export:** Lists of high-risk customers can be exported for customer success outreach teams.

4. Authentication System

The system uses a session-based authentication mechanism:

- **Credentials:** Controlled via hardcoded mock credentials (admin@gmail.com / admin@123).
- **Persistence:** Employs URL query parameters (?logged_in=true) to maintain the session even if the browser is refreshed.
- **Access Control:** The entire UI is gated; unauthorized users are forcibly redirected to the login card.

5. Data Processing & Preprocessing

Handled by the ChurnDataProcessor class:

- **Missing Values:** Uses SimpleImputer with 'median' strategy for numbers and 'most_frequent' for text.
- **Scaling:** Features are normalized using StandardScaler to ensure the model isn't biased by large charge values.
- **Encoding:** OneHotEncoder converts categorical text (like Payment Method) into numerical vectors.
- **Feature Names:** The system tracks feature names through transformations to maintain explainability.

- **Limitation:** Preprocessing is currently "static" and assumes a specific set of core columns exist.

6. Machine Learning Model

The intelligence core utilizes an ensemble approach:

- **Architecture:** The system trains multiple models (Logistic Regression, Random Forest, XGBoost) and selects the "Best" based on the **F1-Score** (balancing precision and recall).
- **Logic:** It uses **SMOTE** (Synthetic Minority Over-sampling Technique) to handle class imbalance, ensuring the model doesn't ignore churners who are typically the minority in a dataset.
- **Explainability:** Integrated with **SHAP** (SHapley Additive exPlanations) to tell the user *why* a customer is high risk (e.g., "High monthly charges are driving this risk").

7. Dashboard & Analytics

- **KPI Cards:** Display Total Customers, Average Churn Rate, and Monthly Revenue.
- **Risk Gauge:** A visual needle indicating the real-time AI prediction for individual profiles.
- **Impact Bar Charts:** Identifies which features (Contract Type, Tenure, etc.) are the strongest drivers of churn across the entire base.
- **Pie Charts:** Visualizes the distribution of payment methods and contract segments.
- **High-Risk Table:** A prioritized list of customers predicted to leave, enabling immediate business action.

8. Business Value & Impact

ChurnPredict shifts the business from **Reactive to Proactive**:

- **Revenue Protection:** Saving just 5% of at-risk customers can lead to a 25-95% increase in profits.

- **Targeted Marketing:** Instead of broad discounts, companies can target only those with a >70% churn probability.
- **Strategic Insights:** Identifying that "Month-to-Month" contracts are the highest risk allows the business to pivot toward Annual plan incentives.

9. Issues & Limitations (Audit Findings)

1. **Authentication:** Lack of a real database backend (e.g., Supabase/PostgreSQL) makes it unsuitable for multiple users.
2. **Data Dependency:** The dashboard heavily relies on exact column name matches or successful mapping; unexpected CSV formats may cause errors.
3. **UI Isolation:** The Streamlit environment lacks some advanced interactive filtering found in dedicated BI tools like Tableau.
4. **Local Storage:** Prediction logs and model artifacts are stored on the local filesystem, which doesn't scale for cloud deployment without shared storage.

10. Recommendations & Improvements

1. **Backend Integration:** Connect to a real Auth provider (Firebase/Supabase) and a persistent SQL database.
2. **Advanced AI:** Implement Time-Series forecasting to predict *when* a customer will churn, not just *if*.
3. **Real-Time Alerts:** Integrate with Slack or Email to notify Account Managers the moment a "High Risk" customer is identified.
4. **Multi-Tenant Support:** Allow different departments to manage their own datasets and models.

11. How It Helps in Data Analysis

- **Pattern Recognition:** Analysts can see at a glance if churn is concentrated in specific payment methods or contract types.

- **What-If Analysis:** The "Intelligence Panel" allows analysts to simulate customer profiles (e.g., "If we move this user to a 1-year contract, how does their risk change?") to validate retention strategies.
- **Data-Driven Culture:** Moves the conversation from "I think people are leaving" to "The AI shows a 75% risk for this segment.

12. Machine Learning Models & Libraries Used

Core Libraries

The intelligence engine is built on the industry-standard Python data science stack:

- **scikit-learn:** The primary framework used for model training, data splitting, and performance evaluation.
- **pandas:** Utilized for high-performance data manipulation, CSV ingestion, and feature mapping.
- **numpy:** Provides the foundation for all numerical operations and vector transformations.
- **XGBoost:** Used for advanced Gradient Boosting, often providing the highest predictive power for tabular churn data.
- **joblib:** Handles the serialization and loading of trained models and preprocessing artifacts.
- **SHAP:** Powers the "Risk Insight" by explaining the contribution of each feature to the final prediction.

Model Analysis

The system evaluates three distinct algorithms to ensure the highest accuracy:

1. **Logistic Regression:**
    - *Role:* Used as a high-interpretability baseline.
    - *Strength:* Excellent for understanding linear relationships and providing a fast, stable starting point.

2. **Random Forest Classifier:**
    - *Role:* An ensemble model that combines multiple decision trees.
    - *Strength:* Handles non-linear relationships and interactions between features (e.g., high charges combined with low tenure) much better than linear models.
3. **XGBoost (Extreme Gradient Boosting):**
    - *Role:* A state-of-the-art boosting algorithm.
    - *Strength:* Highly optimized for speed and performance; typically wins over other models in tabular classification tasks.

Training & Prediction Process

- **Handling Imbalance:** The system uses **SMOTE** to generate synthetic data for the minority class (churners), ensuring the models don't become biased toward "No Churn."
- **Optimization:** Uses **RandomizedSearchCV** to automatically tune hyperparameters (like tree depth and learning rate) to find the best possible configuration.
- **Metric Selection:** Models are compared using the **F1-Score**, which is critical for churn as it balances the need to find all churners (**Recall**) while avoiding false alarms (**Precision**).

Evaluation Metrics

- **Accuracy:** Overall percentage of correct predictions.
- **Precision:** How many predicted churners actually left (Avoiding false positives).
- **Recall:** How many actual churners the system successfully caught (Avoiding false negatives).
- **F1-Score:** The harmonic mean of Precision and Recall; this is our **Primary KPI** for model selection.